Technical Specification: Randomization, Blinding, and Withdrawal Engine

Target Audience: Software Developers / Architects **Core Architecture:** Multi-Study Configuration Engine (Single Platform, Multiple Behaviors)

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1. Architectural Strategy: Study-Type Configuration Engine

The platform must implement a **Capability Matrix** driven by the `study_type`. Instead of hardcoded logic, use a registry to define features per study type.

Supported Study Types

```
type StudyType = 'RCT' | 'PROSPECTIVE' | 'RETROSPECTIVE' | 'CROSS_SECTIONA
```

Capability Interface

```
interface StudyCapabilities {
  randomization: boolean | 'optional';
  allocationConcealment: boolean | 'conditional';
  blinding: boolean | 'optional';
  emergencyUnblinding: boolean;
  finalUnblinding: boolean;
  statisticianMasking: boolean;
  withdrawal: boolean;
  lossToFollowUp: boolean;
}
```

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2. Randomization Engine (RCT Focus)

🚫 Implementation Constraints

- **NO `Math.random()`:** Use a Cryptographically Secure Pseudo-Random Number Generator (CSPRNG).
- **Immutability:** Once a randomization sequence is activated, it must be immutable. Any changes require a new version with a full audit trail.

Supported Algorithms

- Simple Randomization
- Permuted Block (Fixed & Variable)
- Stratified Randomization (Block-based)
- *Future-proofing:* Cluster, Minimization, Adaptive.

Sequence Schema

```
TABLE randomization_sequences (
    id UUID PRIMARY KEY,
    study_id UUID REFERENCES studies(id),
    method ENUM('SIMPLE', 'BLOCK', 'STRATIFIED'),
    block_size INT,
    stratification_factors JSONB,
    status ENUM('DRAFT', 'ACTIVE', 'LOCKED'),
    version INT DEFAULT 1,
    created_at TIMESTAMP
);
```

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3. Blinding & Allocation Concealment

Multi-Level Blinding Model

Blinding must be enforced **Server-Side**. The API must filter sensitive fields based on the user's role and the study's blinding configuration.

Role	Visibility (Blinded)	Visibility (Unblinded)
Participant	Masked (ARM_A)	Actual Treatment
Outcome Assessor	Masked (ARM_A)	Actual Treatment
Statistician	Masked (ARM_A)	Actual Treatment
PI / Admin	Actual Treatment	Actual Treatment

Allocation Concealment Methods

- Central Web Randomization (IVRS/IWRS style)
- Sequentially Numbered Opaque Sealed Envelopes (SNOSE)
- Pharmacy-controlled Allocation

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4. Unblinding Workflows

Emergency Unblinding (Individual Level)

- **Trigger:** Clinical emergency.
- **Requirement:** Reason + Clinical Justification.
- **Outcome:** Reveal allocation for **one participant only**.
- **Audit:** Mandatory log entry for request, approval, and reveal.

Final Unblinding (Study Level)

Locked until the following sequence is completed:

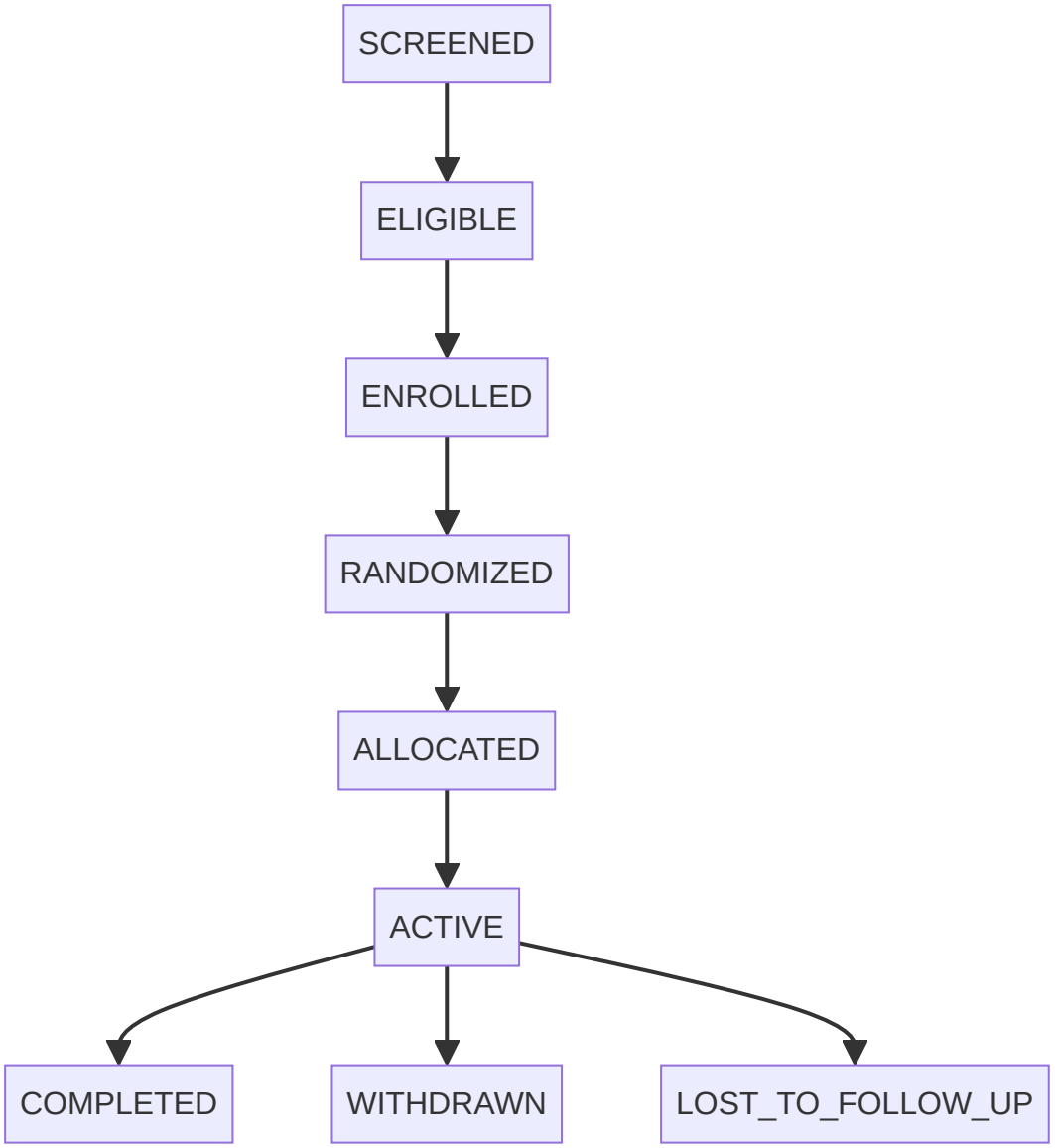
1. Recruitment & Follow-up Complete
2. Data Cleaning & Freeze
3. Database Lock
4. Blinded Statistical Analysis Approved
5. Final Unblinding Triggered

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5. Participant State Machine & Withdrawal

State Transitions

Participants must move through a defined lifecycle. **Status History is mandatory.**



Withdrawal Policy

- **NO DELETION:** Withdrawn participants must remain in the database for ITT (Intention-to-Treat) analysis.
- **Metadata:** Record `withdrawal_date`, `reason` (Participant Decision, Safety, Protocol Violation, etc.), and `requested_by`.

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6. Security & Audit Trail

- **RBAC:** Role-Based Access Control integrated with Blinding Policies.
- **Server-Side Filtering:** Ensure sensitive data (Actual Arm) never leaves the server for blinded roles.
- **Audit Logs:** Immutable logs for all sensitive actions (Randomization, Unblinding, Status Changes).

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7. Capability Matrix (Quick Reference)

Feature	RCT	Prospective	Retrospective	Cross-Sectional
Randomization	YES	Optional	NO	NO
Blinding	YES	Optional	Data Masking	Optional
Emergency Unblinding	YES	Conditional	NO	NO
Withdrawal	YES	YES	Data Withdrawal	YES
Audit Trail	Mandatory	Mandatory	Mandatory	Mandatory